

The Foundation

”

**The Central Dogma. This states that
once 'information' has passed into
protein it cannot get out again.**

”

— Francis Crick, 1958

The Three Pillars



Replication

The process by which a cell duplicates its DNA prior to cell division, ensuring vital information is passed on.



Transcription

The synthesis of an RNA copy from a segment of DNA, taking the instructions out of the protected nucleus.



Translation

The decoding of the mRNA sequence by a ribosome complex to build a highly specific protein molecule.

The Master Blueprint: DNA

The Double Helix Structure

Deoxyribonucleic acid (DNA) is a robust, double-stranded molecule responsible for the long-term storage of the genetic code.

The two strands are antiparallel and are firmly held together by hydrogen bonds between complementary nucleotide base pairs: Adenine (A) pairs with Thymine (T), and Cytosine (C) pairs with Guanine (G).



DNA versus RNA

Deoxyribonucleic Acid (DNA)

- **Structure:** Stable, double-stranded helix.
- **Sugar Base:** Deoxyribose.
- **Nucleobases:** Adenine, Thymine, Cytosine, Guanine.
- **Function:** Long-term, secure storage of genetic information within the nucleus.

Ribonucleic Acid (RNA)

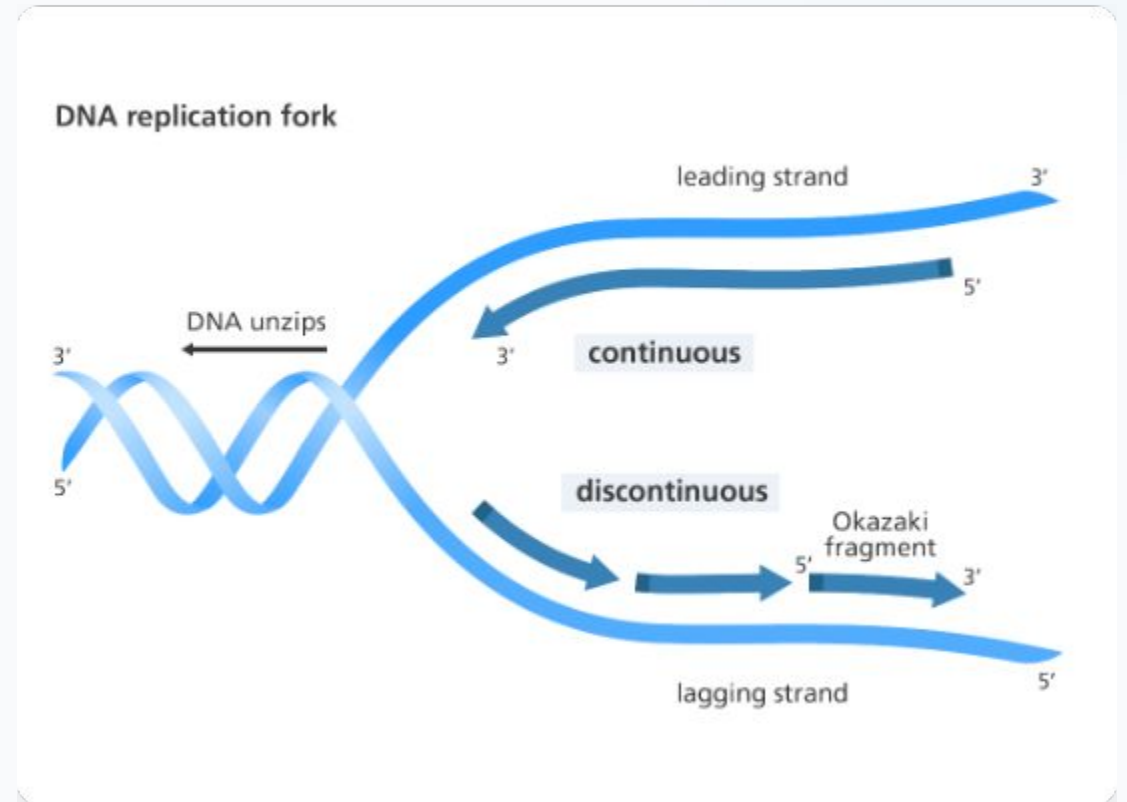
- **Structure:** Typically single-stranded and flexible.
- **Sugar Base:** Ribose.
- **Nucleobases:** Adenine, Uracil (replaces T), Cytosine, Guanine.
- **Function:** Transfers genetic instructions and actively synthesizes proteins.

The Replication Fork

DNA replication is a **semi-conservative** process. This means each newly formed double helix contains one original parent strand and one newly synthesized strand.

The origin of replication is the site where the double helix unwinds, forming a characteristic Y-shaped structure known as the "replication fork."

Because DNA strands run antiparallel, enzymatic synthesis occurs continuously on the **leading strand**, but discontinuously in small sections (Okazaki fragments) on the **lagging strand**.



Helicase



Unzips the DNA double helix by breaking the hydrogen bonds between base pairs, creating the replication fork.

Primase



Synthesizes a short RNA primer, providing a starting point for DNA synthesis.

DNA: ACGATCAGCTAGCA...

Primer: ?????

DNA Polymerase



Binds to the primer and adds complementary nucleotides to build the new, growing DNA strand.

DNA: ACGATCAGCTAGCA...

Primer: ????? ???????????...

Ligase



Seals the gaps between Okazaki fragments on the lagging strand, ensuring a continuous sugar-phosphate backbone.

Helicase



Unzips the DNA double helix by breaking the hydrogen bonds between base pairs, creating the replication fork.

Primase



Synthesizes a short RNA primer, providing a starting point for DNA synthesis.

DNA: ACGATCAGCTAGCA...

Primer: UGCUA

DNA Polymerase



Binds to the primer and adds complementary nucleotides to build the new, growing DNA strand.

DNA: ACGATCAGCTAGCA...

Primer: UGCUAGTCGATCGT...

Ligase

Seals the gaps between Okazaki fragments on the lagging strand, ensuring a continuous sugar-phosphate backbone.

Stages of Transcription



Initiation

RNA Polymerase recognizes and binds to a specific promoter sequence on the DNA, unwinding a small portion of the helix to begin synthesis.



Elongation

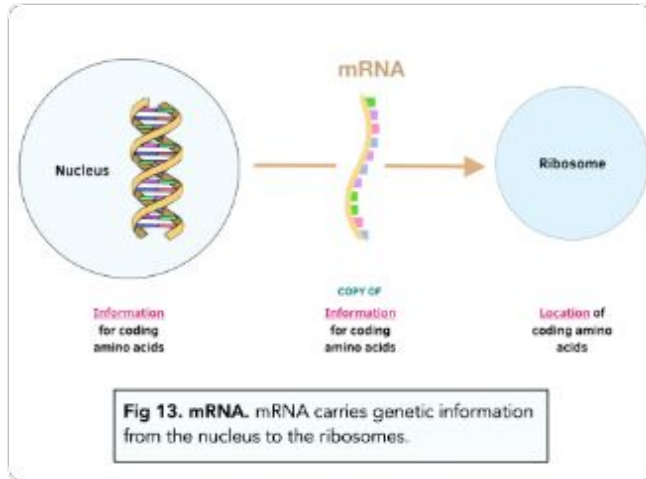
The polymerase enzyme moves along the template strand, accurately assembling the RNA transcript using complementary base pairing.



Termination

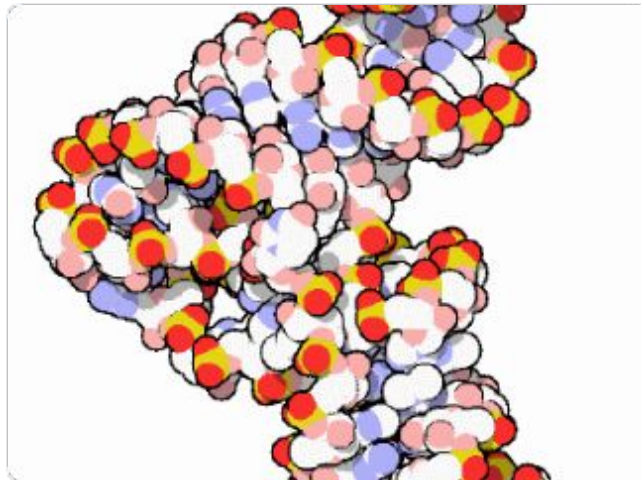
Upon reaching a terminator sequence, the polymerase detaches from the template, and the newly formed RNA strand is released.

The Three Types of RNA



Messenger (mRNA)

Carries the vital genetic instructions copied from DNA out to the ribosomes in the cytoplasm.



Transfer (tRNA)

Matches specific individual amino acids to the corresponding mRNA codons during protein synthesis.



Ribosomal (rRNA)

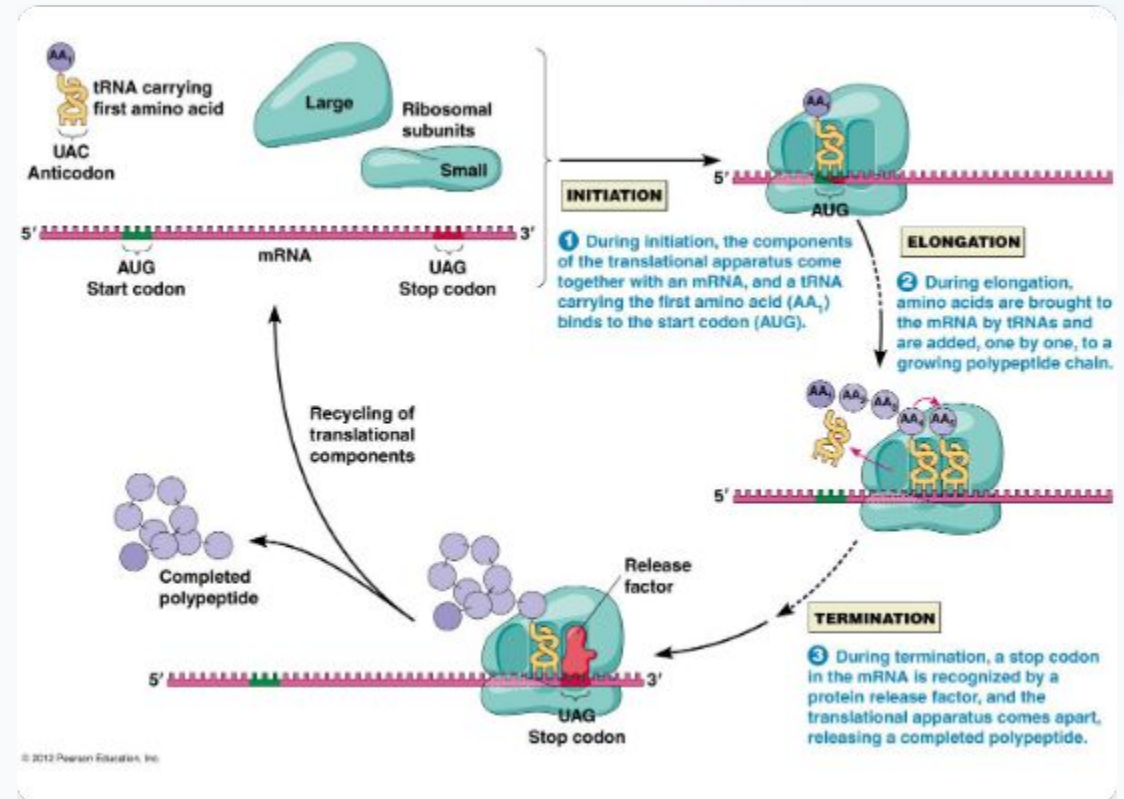
Forms the core physical framework and provides the crucial catalytic activity of the ribosome.

Decoding the Message

Translation is the process where ribosomes synthesize proteins using the mature mRNA transcript.

The ribosome complex carefully assembles around the target mRNA. The very first tRNA, carrying Methionine, is attached at the start codon to initiate the process.

The ribosome then shifts along the mRNA, matching incoming tRNA anticodons to the mRNA codons. Each tRNA efficiently adds its specific amino acid to the growing polypeptide chain via strong peptide bonds.



First letter

Second letter

	U	C	A	G	
U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
A	AUU } Ile AUC } AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
G	GUU } Val GUC } GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

Third letter

	AMINO ACID		
Nonpolar, aliphatic R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{H} \end{array}$ Glycine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_3 \end{array}$ Alanine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH} \\ \quad \\ \text{CH}_3 \text{CH}_3 \end{array}$ Valine
	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH} \\ \quad \\ \text{CH}_3 \text{CH}_3 \end{array}$ Leucine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{S} \\ \\ \text{CH}_3 \end{array}$ Methionine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_2 \\ \\ \text{CH}_3 \end{array}$ Isoleucine
Polar, uncharged R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ Serine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_3 \end{array}$ Threonine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{SH} \end{array}$ Cysteine
	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_2\text{N}^+-\text{C}-\text{H} \\ \quad \\ \text{H}_2\text{C} \quad \text{CH}_2 \end{array}$ Proline	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C} \\ \quad \\ \text{H}_2\text{N} \quad \text{O} \end{array}$ Asparagine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{C} \\ \quad \\ \text{H}_2\text{N} \quad \text{O} \end{array}$ Glutamine

	AMINO ACID		
Positively charged R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH}_3^+ \end{array}$ Lysine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH} \\ \\ \text{C}=\text{NH}_2^+ \\ \\ \text{NH}_2 \end{array}$ Arginine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}-\text{NH}^+ \\ \quad \\ \text{H} \quad \text{CH} \\ \quad \\ \text{H} \quad \text{NH}^+ \end{array}$ Histidine
Negatively charged R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{COO}^- \end{array}$ Aspartate	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{COO}^- \end{array}$ Glutamate	
Nonpolar, aromatic R groups	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_5 \end{array}$ Phenylalanine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_4 \\ \\ \text{OH} \end{array}$ Tyrosine	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_8\text{H}_6\text{N} \end{array}$ Tryptophan

Structures of the 20 amino acids found in proteins are shown. Each amino acid is composed of an amino group (NH_3^+), a carboxyl group (COO^-), and a side chain (blue). The side chain may be

<https://oertx.highered.texas.gov/courseware/lesson/1682/student/?section=6>

The Genetic Code (Codons)

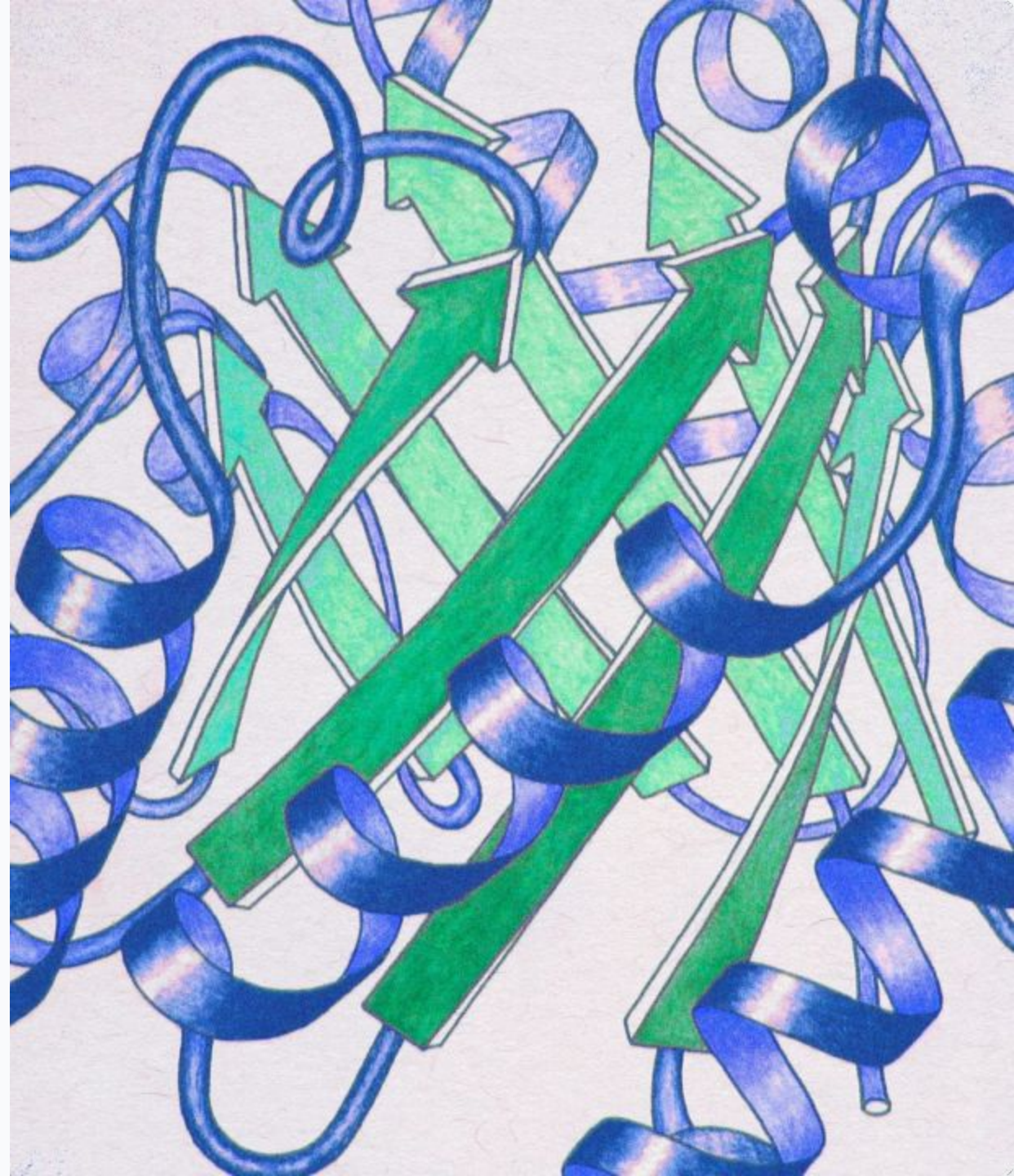
Codon Type	RNA Sequence	Amino Acid Assigned	Biological Function
Start Codon	AUG	Methionine (Met)	Strictly initiates the translation process.
Stop Codons	UAA, UAG, UGA	None	Signals the ribosome to terminate translation.
Standard Codons	61 Variations	20 Different Acids	Specifies the exact sequence of the polypeptide chain.

The Final Product: Proteins

Form Defines Function

Once the long polypeptide chain is fully synthesized, it must immediately fold into a highly specific three-dimensional shape to become a functional protein.

This unique conformation is entirely driven by the complex chemical interactions of the individual amino acid side chains and is absolutely vital for the protein's biological activity within the organism.



Protein Folding Levels



Primary Structure

The linear, one-dimensional sequence of amino acids in the polypeptide chain, dictated directly by the mRNA sequence.



Secondary Structure

Local folding patterns within the chain, primarily alpha-helices and beta-pleated sheets, stabilized by hydrogen bonds.



Tertiary Structure

The overall, complex three-dimensional shape of a single protein molecule, dictated by distant side-chain interactions.

1. Make a DNA Replication of the sequence below

ACATCACGGCAGG (DNA)

???????????? (DNA)

2. Make a transcription of the sequence below

ATGATCACGGCAGG (DNA)

???????????? (RNA)

3. Make a translation of the sequence below

AUG AUC AUG GUA GGG (RNA)

??? ??? ??? ??? ??? (Protein/Peptide)

1. Make a DNA Replication of the sequence below

ACATCACGGCAGG (DNA)

TGTAGTGCCGTCC (DNA)

2. Make a transcription of the sequence below

ATGATCACGGCAGG (DNA)

UACUAGUGCCGUCC (RNA)

3. Make a translation of the sequence below

AUG AUC AUG GUA GGG (RNA)

Met Ile Met Val Gly (Protein/Peptide)

Fidelity of Information Flow

10⁻⁹

ERROR RATE

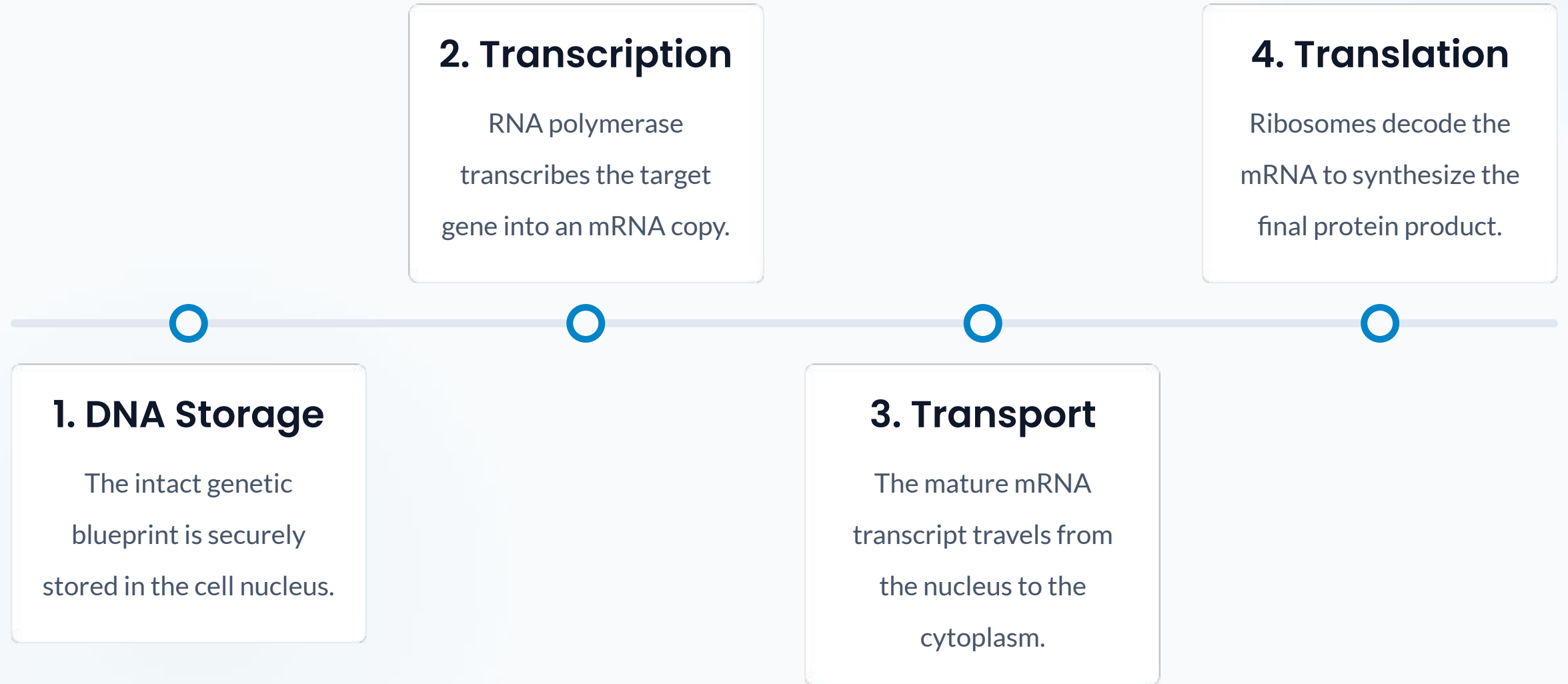
Exceptional Precision

The Central Dogma relies on extreme precision to function.

Thanks to the sophisticated proofreading abilities of enzymes like DNA polymerase, errors in copying the genome are exceptionally rare.

This incredible fidelity (approximately one mistake per billion nucleotides) ensures long-term genetic stability across generations.

Summary: Information Flow



DNA: The Blueprint

- **Structure:** Double helix, antiparallel strands, deoxyribose-phosphate backbone.
- **Sequence Example:**
 - $5' - ATGCGTACCGGTTCA - 3'$
 - $3' - TACGCATGGCCAAGT - 5'$
- **Key Feature:** Focus on the specificity of base pairing (Watson-Crick pairing) and the stability provided by hydrogen bonding.

RNA: The Messenger

- **Structure:** Single-stranded, contains Ribose (2'-OH group), Uracil replaces Thymine.
- **Sequence Example:**
 - $5' - AUGCGUACCGGUUCA - 3'$
- **Key Feature:** Highlight how the 2'-OH group makes RNA more chemically reactive and less stable than DNA, suitable for transient signaling.

Protein: The Functional Product

- **Structure:** Polypeptide chain (amino acid sequence), stabilized by peptide bonds, folds into complex 3D tertiary structures (alpha-helices and beta-sheets).
- **Sequence Example (Three-Letter/Single-Letter Code):**
 - *Met – Arg – Thr – Gly – Ser*
 - *M – R – T – G – S*
- **Key Feature:** The properties of the R-group (side chain) of each amino acid determine the final protein folding and function.

Molecule	Sugar	Bases	Typical Structure
DNA	Deoxyribose	A, T, C, G	Double Helix
RNA	Ribose	A, U, C, G	Single-Stranded
Protein	N/A	Amino Acids	Folded Globular/Fibrous

Image Sources



https://media.istockphoto.com/id/2227323055/photo/abstract-3d-render-of-a-dna-double-helix-made-of-water-symbolizing-genetics-biotechnology-and.jpg?s=612x612&w=0&k=20&c=1LUGzYc82_ho8JEW87I8VMGw8MuOTjxL026vLnln3Y=

Source: www.istockphoto.com



https://cdn.zmescience.com/wp-content/uploads/2018/06/dna_replication_yourgenome.png

Source: www.zmescience.com



<https://studymind.co.uk/wp-content/uploads/2022/03/Screen-Shot-2022-03-31-at-7.53.40-PM.png>

Source: studymind.co.uk



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Image Sources

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